

Ques. WHAT IS THE DIFFERENCE BETWEEN BLUETOOTH 2.X AND BLUETOOTH 4.X?

A. Bansal

Ans. Bluetooth is a wireless technology standard for exchanging data over short distances from fixed and mobile devices and building personal area networks. It uses short-wavelength UHF radio waves in the industrial, scientific and medical radio (ISM) band from 2.4 to 2.485GHz. Bluetooth can connect several devices, overcoming the problems of synchronisation.

Over the years, various versions of Bluetooth have been evolving since it was conceived in 1994. This includes Bluetooth 2.x (for example, v2.0, v2.1, v3.0, v4.0) and Bluetooth 4.x (for example, v4.0, v4.1, v4.2).

Various Bluetooth versions have different speeds and data rate requirements. Some specific optional features and differences between Bluetooth 2.x and 4.x are explained below.

Bluetooth 2.x. This version introduces the optional enhanced data rate (EDR) capability. The most popular variant, Bluetooth 2.1, makes it easier to pair with different devices, even from different manufacturers, and also increases the reliability of the pairing process. Bluetooth 2.x devices with basic rate (basic rate supports a bit rate of 1Mbps) plus EDR have a theoretical maximum data rate up to 3Mbps.

Bluetooth 4.x. This version introduces support for collecting data from very-low-power devices. The main intent of this feature, called Bluetooth Low Energy (also, BLE or LE), is to aggregate data from various sensors like heart rate monitors and thermometers. The commercial name of LE feature is Bluetooth Smart. This version supports basic rate plus three optional features, namely, EDR, high-speed and LE.

The newer version has improved upon power consumption, data rate, security, pairing reliability, ease and speed, among others.

Q2. HOW DO SMARTPHONES GET VIBRATIONS?

Nidhi K.

A2. We are all familiar with a smartphone's vibrations, whether it is for alerting us to a phone call or setting the phone on vibration mode for other applications.

Vibrations are caused by a small component like the one shown in the picture. Haptic refers to the sense of touch, which is used to provide feedback to electronic devices such as smartphones and tablets through vibrations.

There are two types of motors to set off those vibrations: eccentric rotating motor (ERM) and linear resonant actuator (LRA). Whether the vibrations are simple or sophisticated, the process begins with either a touch or a signal from the software or hardware.

An electrical signal from hardware, such as an incoming phone call or the push of a button, goes to a microprocessor dedicated to haptic creations. You can have a library of vibrations for different occasions or you can program these to create a unique touch sensation.

The microprocessor decides what type of vibrations are needed, but since it does not have enough power to create the vibrations, it hands over the job to a haptic driver. Drivers might be included in some processors.

Using the control signal from the

processor as a guide, the haptic driver sends current, which rapidly changes in strength to one of two types of haptic actuators/motors, causing vibrations.

Q3. HOW DOES A TRANSISTOR WORK IN THE WORLD OF ELECTRONICS?

Riya T.

A3. The most important invention of the 20th century is a transistor switch, whose functions are similar to a wall switch. It lets electricity flow through it or opens to block the current. The difference is that the transistor can be made exceedingly small. Due to its small size, you can make a microprocessor with millions of transistors. A transistor has the ability to turn on and off so rapidly that the number of times it closes or opens is measured in millions of a second.

Functions of a transistor include amplification and switching. For example, weak signals coming from a microphone can be strengthened using power amplifying transistor(s), so that it can be heard through a loudspeaker. Without changing the waveform of the input signal, only voltage or current of the signal is amplified.

While this is the case of analogue signals, in the case of digital signals used in computers, the transistor plays the role of a switch, changing the state between 0 and 1. Integrated circuits and very-large-scale-integrations are basically collections of transistors. In this electronics age, these small transistor devices are directly or indirectly part of many innovations and scientific wonders.



Mobile phone flat-coin vibration motor

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